

# Distributions of Cells and Neurons across the Cortical Sheet in Old World Macaques

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## Key Words

Brain size · Cell density · Glia · Neocortex · Primates · Visual cortex

## Abstract

According to previous research, cell and neuron densities vary across neocortex in a similar manner across primate taxa. Here, we provide a more extensive examination of this effect in macaque monkeys. We separated neocortex from the underlying white matter in 4 macaque monkey hemispheres (1 *Macaca nemestrina*, 2 *Macaca radiata*, and 1 *Macaca mulatta*), manually flattened the neocortex, and divided it into smaller tissue pieces for analysis. The number of cells and neurons were determined for each piece across the cortical sheet using flow cytometry. Primary visual cortex had the most densely packed neurons and primary motor cortex had the least densely packed neurons. With respect to differences in brain size between cases, there was little variability in the total cell and neuron numbers within specific areas, and overall trends were similar to what has been previously described in Old World baboons and other primates. The average hemispheric total cell number per hemisphere ranged from 2.9 to 3.7 billion, while the average total

neuron number ranged from 1.3 to 1.7 billion neurons. The visual cortex neuron densities were predictably higher, ranging from 18.2 to 34.7 million neurons/cm<sup>2</sup> in macaques, in comparison to a range of 9.3–17.7 million neurons/cm<sup>2</sup> across cortex as a whole. The results support other evidence that neuron surface densities vary across the cortical sheet in a predictable pattern within and across primate taxa.

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## Introduction

This study is part of our ongoing efforts to characterize regional cell and neuron populations in primate neocortex. The primate cortex is divided into a large number of functionally specialized areas, which vary by neuron types, laminar patterns, and physiological properties [Sherwood and Hof, 2007; de Sousa et al., 2010]. The adult cortex of non-hominin primates has a characteristic uneven distribution of cells and neurons across its extent that has been recently detailed in a small number of individuals across a range of primate taxa (one macaque, three baboons, and one galago) [Collins et al., 2010a; Young et al., 2013a, b]. The uneven cell and neuron dis-

tribution is indicative of the variable processing roles and capabilities of different areas of the cerebral cortex.

Typically, studies of cell and neuron numbers in cortex only report the total numbers in the cerebral cortex [Pakkenberg and Gundersen, 1997; Christensen et al., 2007; Herculano-Houzel et al., 2008]. Other studies have provided counts for a few locations across the cortical sheet, in support of the conclusion that cell densities are rather uniform across cortex, with the exception of a twofold increase in primary visual cortex of primates [Rockel et al., 1980; Carlo and Stevens, 2013]. In contrast, studies in rodents [Skoglund and Pascher, 1996; Herculano-Houzel et al., 2013], cats [Beaulieu and Colonnier, 1989], and primates [Collins et al., 2010a, 2016; Young et al., 2013a, b] have revealed consistent quantitative differences between cortical areas. Previous efforts in primates have confirmed that primary visual cortex (V1) and other secondary visual areas contain higher than average neuron numbers under a fixed amount of cortical surface area, while primary motor cortex (M1) contains lower than average neuron numbers. Such areal differences likely reflect differences in laminar composition, as motor areas have more widely distributed, large pyramidal cells in layers 3 and 5 and a minimal layer 4, while primary sensory areas such as V1 tend to contain a wide, neuron-dense layer 4 packed with small granule cells.

The most detailed studies of the densities of neurons across the cortical sheet involve estimates made from many locations, most recently 742 pieces of cortex from a hemisphere of a chimpanzee [Collins et al., 2016], and previous counts from a single macaque, three baboons (and two epileptic baboons), one New World squirrel monkey, and two galagos [Collins et al., 2010a; Young et al., 2013b]. We extend these observations here to examine neuron surface density in detail across three species of Old World macaque monkeys. The genus *Macaca* is extremely diverse, with an estimated 22 different species currently recognized, and the earliest fossil records and molecular data point to their divergence from other Papionini species around 10 million years ago, which suggests a relatively fast-paced spread within the *Macaca* radiation [Delson et al., 2000; Raaum et al., 2005]. The genus *Macaca* is traditionally further subdivided into four species groups (*silenus-sylvanus*, *fascicularis*, *sinica*, and *arctoides*), and while all share a common ancestor, it has been reported that there are differences in brain volume between these species groups, between species in the same species group, and between individuals of the same species [Stephan et al., 1981; Allman et al., 1993; Allman, 1994; Hakeem et al., 1996]. The average body weights for

**Table 1.** Summary of experimental cases

| Case     | Species              | Age, years | Sex | Hemi-sphere | Brain weight, g | Brain surface area, cm <sup>2</sup> |
|----------|----------------------|------------|-----|-------------|-----------------|-------------------------------------|
| 12-58 LH | <i>M. radiata</i>    | 7.5        | M   | LH          | 25.0            | 100.6                               |
| 12-58 RH | <i>M. radiata</i>    | 7.5        | M   | RH          | 24.8            | 103.4                               |
| 11-47    | <i>M. nemestrina</i> | 11         | F   | RH          | 36.4            | 151.8                               |
| 10-50    | <i>M. mulatta</i>    | 14.2       | M   | LH          | 33.2            | 152.3                               |
| 9-27     | PHX                  | 12.6       | F   | RH          | 56.3            | 186.0                               |
| 11-31    | PHA                  | 13         | F   | RH          | 50.0            | 234.0                               |

PHA = *P. hamadryas anubis*; PHX = *P. hamadryas anubis/cyncephalus* hybrid. In case 12-58, the frontal eye field and surrounding areas are missing, and in case 10-50, the A-P coordinates could not be obtained.

*Macaca nemestrina* (males: 11.2 kg, females: 6.5 kg) [Fooden, 1975] and *Macaca mulatta* (males: 11 kg, females: 8.8 kg) [Fooden, 1981] are considerably higher than the average body weight for *Macaca radiata* (males: 6.7 kg, females: 3.9 kg) [Schwartz and Kemnitz, 1992]. It is then likely that there are corresponding differences in brain size, the size of the cortical sheet, and, perhaps, numbers and distributions of cortical neurons. The present study serves as a further introduction to the discussion of individual and species differences in cell and neuron neocortical densities.

## Materials and Methods

To evaluate the areal and regional differences in the distributions of cell and neuron numbers in cortex, neocortex from four hemispheres of three adult macaques [1 *M. nemestrina*, 2 *M. radiata* (same animal), and 1 *M. mulatta*] was separated from underlying structures, manually flattened, dissected into small pieces, and processed for their neuronal and nonneuronal cellular content using the flow fractionator method [Collins et al., 2010b].

### Tissue and Cortex Dissection

Experimental procedures were all approved by the Vanderbilt Institutional Animal Care and Use Committee. Cortex samples from four adult macaques were used in this study. Case 11-47 was purchased from the tissue program at the University of Washington National Primate Research Center. Cases 10-50 and 12-58 were obtained after unrelated experiments at Vanderbilt University. Young et al. [2013b] reported data for baboon cases 09-27 and 11-31, and those cell and neuron numbers are referenced and displayed in figures here for comparison. Table 1 summarizes specific details on the species, age, sex, hemisphere, and size specifications of each case. All brains were perfused with 0.9% phosphate-buffered saline (PBS) and 2% paraformaldehyde. Brains that were

obtained outside the Vanderbilt University were shipped overnight in PBS. All brains were bisected into the right (RH) and the left hemisphere (LH); the RH of case 10-50 was used in other studies, the LH of case 11-47 was processed in sections (as described below), and both hemispheres of case 12-58 were used in this study. The cortex was separated from the underlying subcortical structures, the pia mater was removed, and the sulci were opened with forceps to allow for manual flattening, as previously described [Gharbawie et al., 2011; Collins et al., 2016]. Each hemisphere was then placed in a glass dish with PBS, covered with a weighted glass slide, and left for a minimum of 12 h to completely flatten [Gharbawie et al., 2011].

Cortical areas were dissected from the cortical sheet with a scalpel when possible, including primary visual cortex (V1), secondary visual cortex (V2), primary somatosensory cortex (S1), and primary motor cortex (M1). These areas are readily identified when the flat cortex is viewed on a light box to visualize myelin-dense sensory areas that appear darker relative to surrounding cortical areas [Collins et al., 2010a; Campi et al., 2011; Young et al., 2013a]. Areas V2, S1, and M1 could not be confidently obtained from case 10-50 due to lack of photographic documentation during the dissection. Once removed, cortical areas were further dissected into smaller pieces that measured approximately  $5 \times 5$  mm, or smaller, and each piece was assigned to an internal representation (e.g., V1, M1) when possible. With the exception of case 10-50, whose flattened cortex could not be reconstructed, the exact locations of dissection cuts were indicated on a high-resolution photograph of the flattened cortex. Each piece was assigned a number and weighed. The surface area was measured for each piece using ImageJ software (National Institutes of Health). This software was also used to assign an anterior-posterior (A-P) coordinate by generating a centroid measure for the location of each tissue piece. Neuron numbers per square centimeter of surface area versus the A-P locations were plotted, and the data were used in curve estimation regression models to assess the relationships of neuron densities across the A-P dimension with linear, quadratic, cubic, and exponential models. The glial cell/neuron (G/N) ratios versus the A-P locations were also plotted and used in curve-fitting estimations.

#### *Tissue Homogenization for Flow Fractionator*

Each piece was homogenized using a glass Tenbroeck tissue grinder in a solution of sodium citrate and Triton X-100 in distilled water. This method of tissue homogenization produces cells that are completely dissociated with the nuclei remaining intact [Herculano-Houzel and Lent, 2005; Collins et al., 2010b]. Resulting suspensions were stored in 15-ml conical tubes and left to settle for 24–72 h at 4°C. The volume of each sample was adjusted such that they all had the same appearance in terms of their clarity.

#### *Total Number of Cells*

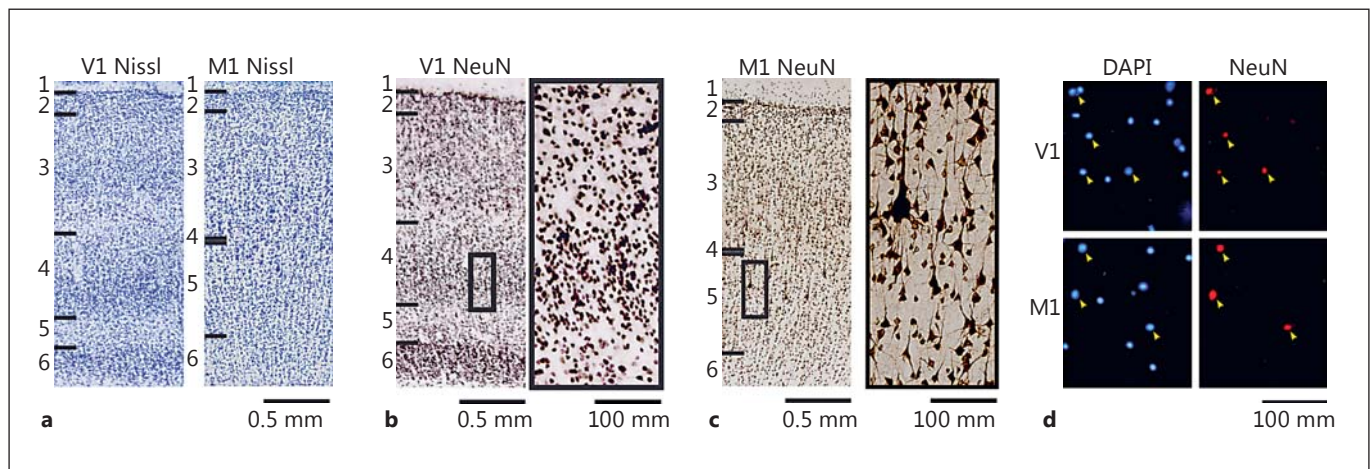
The flow fractionator method was used to estimate the total number of cells per piece from complete counts of samples from the homogenized suspensions, as previously described [Young et al., 2012]. In brief, the suspended samples were well mixed, and a 50- $\mu$ l aliquot was removed and suspended in a 250- $\mu$ l solution of PBS and DAPI (4',6-diamidino-2-phenylindole; 0.5 mg/200 ml). All samples were prepared in duplicate to assess the reliability of each estimate. A 50- $\mu$ l aliquot of CountBright absolute counting beads (Invitrogen) was added to each sample prior to evaluation with the flow cytometer. All samples were evaluated on a Becton

Dickinson LSRII 5-laser flow cytometer using an Xcyte 355-nm laser for DAPI excitation. Samples were well mixed to assure a homogeneous distribution of beads and nuclei, and the flow cytometer evaluated the number of nuclear events that occur coincident with 1,000 CountBright bead events using the flow cytometer. As all samples were prepared in duplicate, a coefficient of variation was calculated for such pairs to detect variation between them that would affect the data. Samples with a coefficient of variation greater than 0.20 were rerun to determine the cell population estimate with greater accuracy. The side scatter (SSC-A) and DAPI parameters were used to selectively place the gate for counting. The flow cytometry expert (D.K.F.) placed the nuclear selection gate and was blind to the sample location in the cortex. All flow cytometry experiments were conducted in the Vanderbilt University Medical Center (VUMC) Flow Cytometry Shared Resource. The isotropic fractionator method [Herculano-Houzel and Lent, 2005] was used on 10% of samples to verify data obtained from the flow fractionator method (2.0 mg DAPI in 200 ml PBS).

#### *Total Number of Neurons*

The flow fractionator method was also used to determine the total number of neurons per cortical piece, as previously described [Collins et al., 2010b]. In brief, the suspension samples were homogenized and a 0.5-ml aliquot was removed and immunolabeled for the neuronal nuclear antigen (NeuN) that is present in neuronal nuclei. All aliquots were antigen retrieved in boric acid (0.2 M, pH = 9.0) at 70°C for 30 min. Samples were washed with PBS and resuspended overnight in PBS and anti-NeuN antibody (1:600; Millipore) at room temperature on a shaker. The following day, samples were washed twice in PBS and resuspended in secondary antibody solution of goat, anti-mouse IgG tagged with Alexa Fluor 647 (AF647; Invitrogen, Inc.) for 1 h. Samples were then washed two more times, stained with DAPI (0.5 mg/200 ml), and then forced through a 35- $\mu$ m mesh filter to remove debris from myelin and cell membranes. All samples were homogenized and evaluated on a Becton Dickinson LSRII 5-laser flow cytometer using the 640-nm laser. The proportion of AF647+ nuclei was determined for 1,000 DAPI+ events. The total number of neurons in each piece was then calculated by multiplying the proportion of neurons by the total cell estimate for that specific sample. The isotropic fractionator method [Herculano-Houzel and Lent, 2005] was again used on 10% of samples to verify data obtained from the flow fractionator method. The NeuN-immunoreactivity (IR) nuclei in suspension in these samples were instead brightly labeled with Alexa Fluor 594 goat anti-mouse IgG secondary antibody (AF594; Invitrogen, Inc.) and not filtered.

The number of other cells in cortex, the majority of which are glial cells [Azevedo et al., 2009; Herculano-Houzel, 2014], was obtained by subtracting the number of neurons from the total number of cells. The G/N ratio [Friede, 1954; Bass et al., 1971; Herculano-Houzel, 2014] was used as a measure to compare cortical areas. A nonparametric Kruskal-Wallis independent-sample test with adjustments for multiple comparisons was used to compare the G/N ratio distributions in V1, V2, S1, and M1. Two additional Kruskal-Wallis independent-sample tests were used to compare the distribution of (1) neuron and (2) cell surface densities in V1, V2, S1, and M1. Statistical significance was set at  $p < 0.05$  for a two-sided test. Data were analyzed using IBM-SPSS software (IBM-SPSS 23).



**Fig. 1.** Nissl- and NeuN-stained tissue in V1 and M1. High-magnification images of NeuN-IR in areas V1 and M1 in *M. nemestrina* (case 11-47). Sections from V1 and M1 were stained for Nissl substance (with thionin) (**a**) to verify the quality of tissue used in this study. Sections from V1 (**b**) and M1 (**c**) were stained with NeuN to verify consistent labeling of neurons with the anti-NeuN antibody. Both the Nissl and NeuN labeling in tissue sections illustrate the typical laminar distribution characteristic of these two cortical areas. A comparison of the NeuN-IR sections in V1 and M1 clearly demonstrate that multiple neuron types are darkly

stained and readily identifiable at high magnification, as V1 has a large layer 4 with densely packed granule cells, layer 4 in M1 is minimal but less densely packed, but large pyramidal cells are found throughout M1 layer 5. V1 and M1 tissue from the LH of the same individual was processed for the flow fractionator with the same anti-NeuN primary antibody (**d**). Nuclei were strongly and brightly labeled with both DAPI and NeuN. There is absence of background or debris in the samples, suggesting that our flow cytometry estimates contain minimal errors.

#### NeuN-IR in Tissue Sections

The intact RH of case 11-47 (*M. nemestrina*) was sectioned into 10 series of 50- $\mu$ m-thick frozen sections. One series of sections was processed for Nissl substance (with thionin) and another series was immunostained for NeuN (1:5,000; Millipore) to verify the tissue and staining quality in this study. Immunohistochemical procedures used have been described in detail previously [Balaram et al., 2014]. Figure 1a shows Nissl-stained intact V1 and M1 sections that exemplify the typical laminar patterns of these cortical areas. Figure 1b, c shows NeuN-IR in the same cortical areas and further confirms the typical laminar distribution of neurons in V1 and M1. The NeuN antibody darkly stained a variety of neuron types that can be verified by their structure, including the small granule neurons in V1 layer 4 (fig. 1b) and large pyramidal neurons in M1 layer 5 (fig. 1c). Figure 1a–c shows that the laminar organization of cortical areas is normal, and there is minimal-to-no evidence of background or staining artifact. This indicates that tissue conditions and preparation in this study were ideal for counting cells. Figure 1d visually confirms anti-NeuN immunolabeling of neuronal nuclei in homogenized neural tissue suspensions of V1 and M1 in the opposite hemisphere of the same individual (case 11-47, LH). The lack of background or artifact staining shows that the homogenization process did not result in broken or damaged nuclei that would interfere with flow cytometry cell and neuron population estimates, and that NeuN-IR nuclei were specifically stained and easily distinguishable from non-neuronal nuclei.

## Results

### Cell and Neuron Numbers and Densities by Case

The total numbers of cells and neurons in the cerebral cortex of macaque monkeys varied minimally between the individuals of the macaque species examined. The total number of cells per cortical hemisphere ranged from 2.9 to 3.7 billion, while the total number of neurons narrowly varied with a range of 1.3–1.7 billion (summarized in table 2). The cell and neuron densities of individual pieces, as well as the boundaries used to define V1, V2, S1, and M1 in each case, are shown in figure 2 for cases where illustration was possible. The S1 region largely corresponds to area 3b, but some of areas 3a, 1, and 2 are likely within the region, and similarly some of area 3a may be in the M1 region. The V2 region does not include all of area V2, but is rather a more conservative estimate; tissue pieces anterior to the V2 boundary line are likely other secondary or tertiary visual areas. The average neuron and cell densities within each cortical area by case are represented in figure 3a and b, respectively, and include available data from two baboon cases [Young et al., 2013b] for comparison. Table 2 summarizes the average and regional densities within cases.

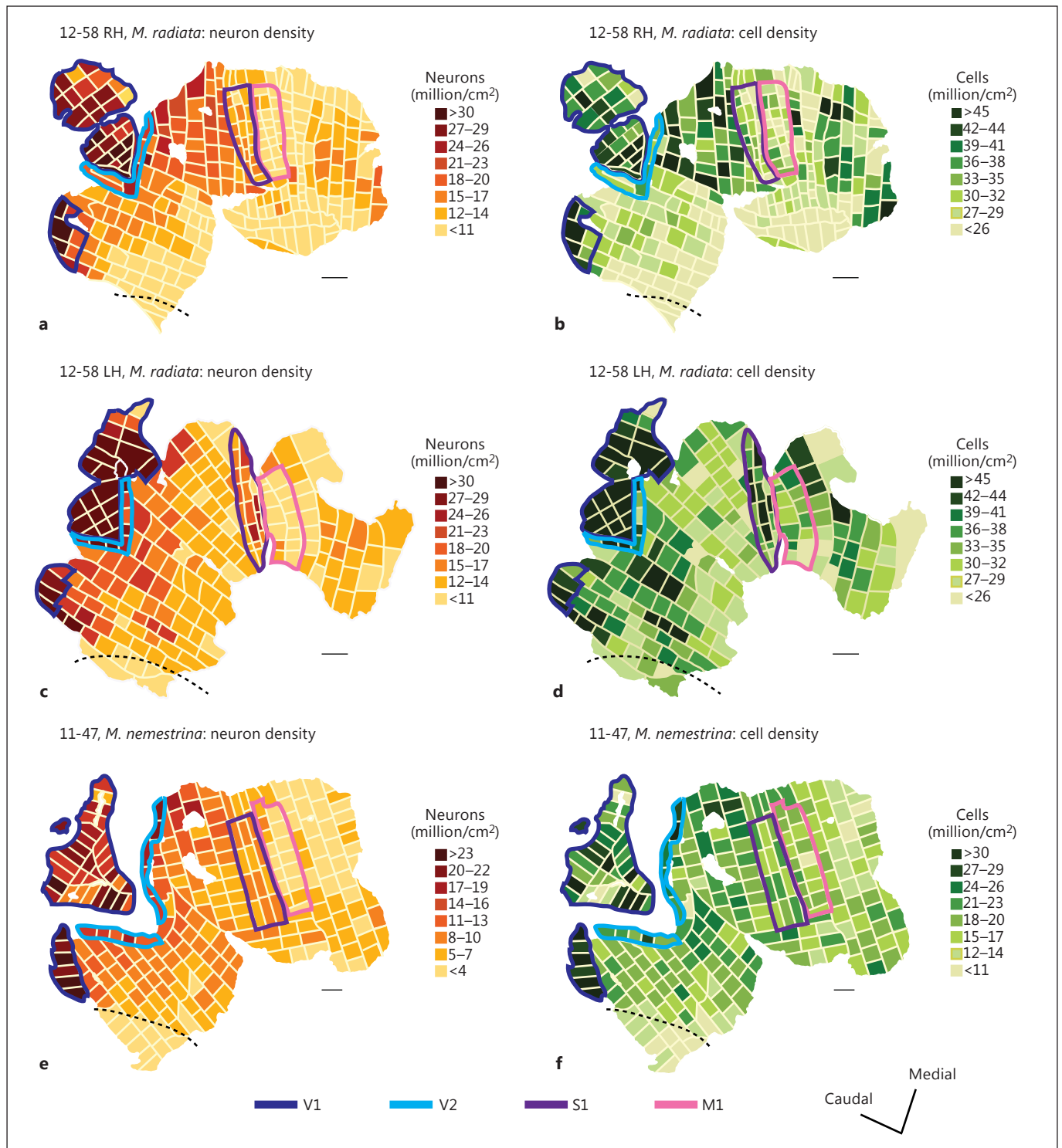
**Table 2.** Summary of cell and neuron data (averaged across all areas and regions of the cortex)

| Case                                                            | Total cells<br>in millions | Total neurons<br>in millions | Total<br>nonneurons<br>in millions | G/N ratio   | Mean cell<br>density in million<br>cells/cm <sup>2</sup> | Mean neuron<br>density in million<br>neurons/cm <sup>2</sup> |
|-----------------------------------------------------------------|----------------------------|------------------------------|------------------------------------|-------------|----------------------------------------------------------|--------------------------------------------------------------|
| Summary of data averaged across all areas and regions of cortex |                            |                              |                                    |             |                                                          |                                                              |
| 12-58 LH                                                        | 3,684 (8.11)               | 1,704 (5.08)                 | 1,980 (4.94)                       | 1.40 (0.79) | 37.6 (9.88)                                              | 17.7 (8.99)                                                  |
| 12-58 RH                                                        | 3,319 (5.27)               | 1,566 (3.52)                 | 1,753 (2.92)                       | 1.44 (1.00) | 32.0 (8.42)                                              | 14.9 (6.84)                                                  |
| 11-47                                                           | 2,960 (3.67)               | 1,344 (3.20)                 | 1,616 (2.56)                       | 1.64 (1.16) | 19.9 (5.23)                                              | 9.26 (5.45)                                                  |
| 10-50                                                           | 2,887 (8.75)               | 1,400 (6.24)                 | 1,487 (4.72)                       | 1.29 (0.69) | 19.2 (5.09)                                              | 9.37 (4.34)                                                  |
| Summary of data averaged across V1                              |                            |                              |                                    |             |                                                          |                                                              |
| 12-58 LH                                                        | 629 (10.66)                | 412 (7.09)                   | 217 (4.23)                         | 0.61 (0.46) | 52.6 (10.87)                                             | 34.7 (9.48)                                                  |
| 12-58 RH                                                        | 641 (5.92)                 | 426 (4.04)                   | 215 (2.27)                         | 0.54 (0.24) | 40.2 (6.71)                                              | 26.9 (6.21)                                                  |
| 11-47                                                           | 506 (5.39)                 | 368 (4.11)                   | 138 (1.56)                         | 0.38 (0.12) | 25.2 (7.39)                                              | 18.3 (5.56)                                                  |
| 10-50                                                           | 384 (10.63)                | 263 (6.76)                   | 121 (4.08)                         | 0.46 (0.07) | 26.6 (2.25)                                              | 18.2 (1.21)                                                  |
| Summary of data averaged across V2                              |                            |                              |                                    |             |                                                          |                                                              |
| 12-58 LH                                                        | 141 (5.73)                 | 96.9 (4.18)                  | 44.1 (1.67)                        | 0.46 (0.07) | 41.1 (7.82)                                              | 28.0 (5.13)                                                  |
| 12-58 RH                                                        | 78.3 (1.69)                | 52.1 (1.45)                  | 26.2 (0.37)                        | 0.51 (0.05) | 32.1 (5.02)                                              | 21.3 (3.85)                                                  |
| 11-47                                                           | 114 (3.08)                 | 77.8 (2.26)                  | 36.2 (0.99)                        | 0.48 (0.08) | 24.7 (4.43)                                              | 16.8 (3.15)                                                  |
| Summary of data averaged across S1                              |                            |                              |                                    |             |                                                          |                                                              |
| 12-58 LH                                                        | 184 (3.19)                 | 78.1 (1.26)                  | 105.9 (2.13)                       | 1.39 (0.37) | 40.2 (10.52)                                             | 17.1 (4.80)                                                  |
| 12-58 RH                                                        | 133 (2.53)                 | 61.0 (1.11)                  | 72.0 (1.49)                        | 1.19 (0.29) | 31.2 (4.88)                                              | 14.3 (2.11)                                                  |
| 11-47                                                           | 129 (1.91)                 | 57.2 (0.95)                  | 71.8 (1.18)                        | 1.27 (0.23) | 19.6 (2.30)                                              | 8.73 (1.35)                                                  |
| Summary of data averaged across M1                              |                            |                              |                                    |             |                                                          |                                                              |
| 12-58 LH                                                        | 202 (4.45)                 | 64.9 (1.55)                  | 137.1 (3.64)                       | 2.25 (0.86) | 32.9 (7.04)                                              | 10.6 (3.23)                                                  |
| 12-58 RH                                                        | 146 (1.80)                 | 50.2 (0.69)                  | 95.8 (1.35)                        | 1.99 (0.65) | 29.4 (4.75)                                              | 10.1 (2.36)                                                  |
| 11-47                                                           | 180 (2.66)                 | 49.7 (0.79)                  | 130.3 (2.16)                       | 2.71 (0.73) | 18.2 (2.60)                                              | 5.06 (1.28)                                                  |

The LH of case 12-58 (excluding the frontal eye field used for another study) had a total surface area of 100.6 cm<sup>2</sup>, a total gray matter mass of 25 g, and contained 3.7 billion cells, of which 1.7 billion were neurons (46%). The cortical cell and neuron densities were 37.6 million cells/cm<sup>2</sup> and 17.7 million neurons/cm<sup>2</sup>. V1 was estimated to contain 629 million cells, of which 412 million were neurons (65%). The cell and neuron densities in V1 were 52.6 million cells/cm<sup>2</sup> and 34.7 million neurons/cm<sup>2</sup>. Anterior to V1, V2 was estimated to contain 141 million cells, of which 97 million were neurons (69%). V2 contained cell and neuron densities of 41.1 million cells/cm<sup>2</sup> and 28 million neurons/cm<sup>2</sup>. S1 was estimated to contain 184 million cells, of which 78 million were neurons (42%). S1 contained cell and neuron densities of 40.2 million cells/cm<sup>2</sup> and 17.1 million neurons/cm<sup>2</sup>. Anterior to S1, M1 was estimated to contain a total of 202 million cells, of which 65 million were neurons (32%). M1 had cell and neuron densities of 32.9 million cells/cm<sup>2</sup> and 10.6 million neurons/cm<sup>2</sup>.

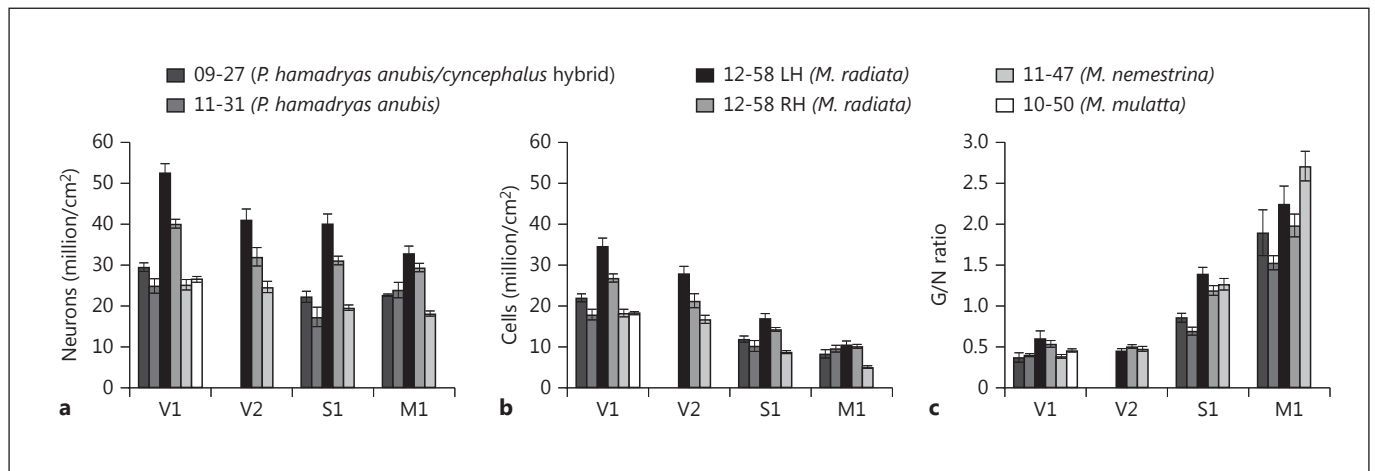
Case 12-58 (RH) had a total surface area of 103.4 cm<sup>2</sup> and a total weight of 24.8 g. The RH was 2.7 cm<sup>2</sup> larger than LH, and weighed 0.8 g more, likely in part due to the

removal of tissue including the frontal eye field from the LH for another study. As listed in table 2, there are slight differences in cell and neuron numbers between the LH and the RH of case 12-58; despite having the slightly higher surface area, the RH contained approximately 365 million less cells and 138 million less neurons than the LH. This may reflect a loss of some of V1 of the RH during the dissection. The RH contained approximately 3.3 billion cells, of which nearly 1.6 billion were neurons (47%). The cell and neuron densities across cortex were approximately 32 million cells/cm<sup>2</sup> and 14.5 million neurons/cm<sup>2</sup>, respectively. V1 was estimated to contain 641 million cells, of which 426 million (67%) were neurons. The cell and neuron densities in V1 were calculated to be 40.2 million cells/cm<sup>2</sup> and 26.9 million neurons/cm<sup>2</sup>. V2 was estimated to contain 78 million cells, of which 52 million were neurons (67%). V2 contained cell and neuron densities of 32.1 million cells/cm<sup>2</sup> and 21.3 million neurons/cm<sup>2</sup>. We estimated that there were a total of 133 million cells and 61 million neurons (46%) in S1. S1 had cell and neuron densities of 31.2 million cells/cm<sup>2</sup> and 14.3 million neurons/cm<sup>2</sup>. M1 was estimated to contain a total of 146 million cells and 50 million neurons (34%). The M1



**Fig. 2.** Distribution of neurons (a, c, e) and cells (b, d, f) across cortex in three macaque hemispheres. Conservative boundaries for V1, V2, S1, and M1 cortical regions are outlined within each hemisphere. Both neuron and cell densities were highest in posterior V1 and lowest in anterior M1, even though the range of estimated values differed between cases. The dashed lines correspond

to tissue pieces that are not part of the neocortex and are likely a part of piriform cortex. Thin white lines represent cuts in the tissue, while larger white spaces within the flattened cortex represent holes in the tissue that were necessary for the flattening process. In some cases, V1 was separated from the rest of cortex to prevent damage during the unfolding process.



**Fig. 3.** The mean neuron (a) and cell (b) surface densities by cortical area per case. Cases 09-27 (*Papio hamadryas anubis/cyncephalus* hybrid) and 11-31 (*P. hamadryas anubis*) are included based on published data [Young et al., 2013b]. c The mean G/N ratio within each cortical area did not vary greatly between cases of the same

genus, but the G/N ratio was considerably lower in the baboon hemispheres across all cortical areas. Most V1 and V2 values were indistinguishable from one another and contained the lowest average G/N ratio. Error bars = SEM.

cell and neuron densities were 29.4 million cells/cm<sup>2</sup> and 10.1 million neurons/cm<sup>2</sup>.

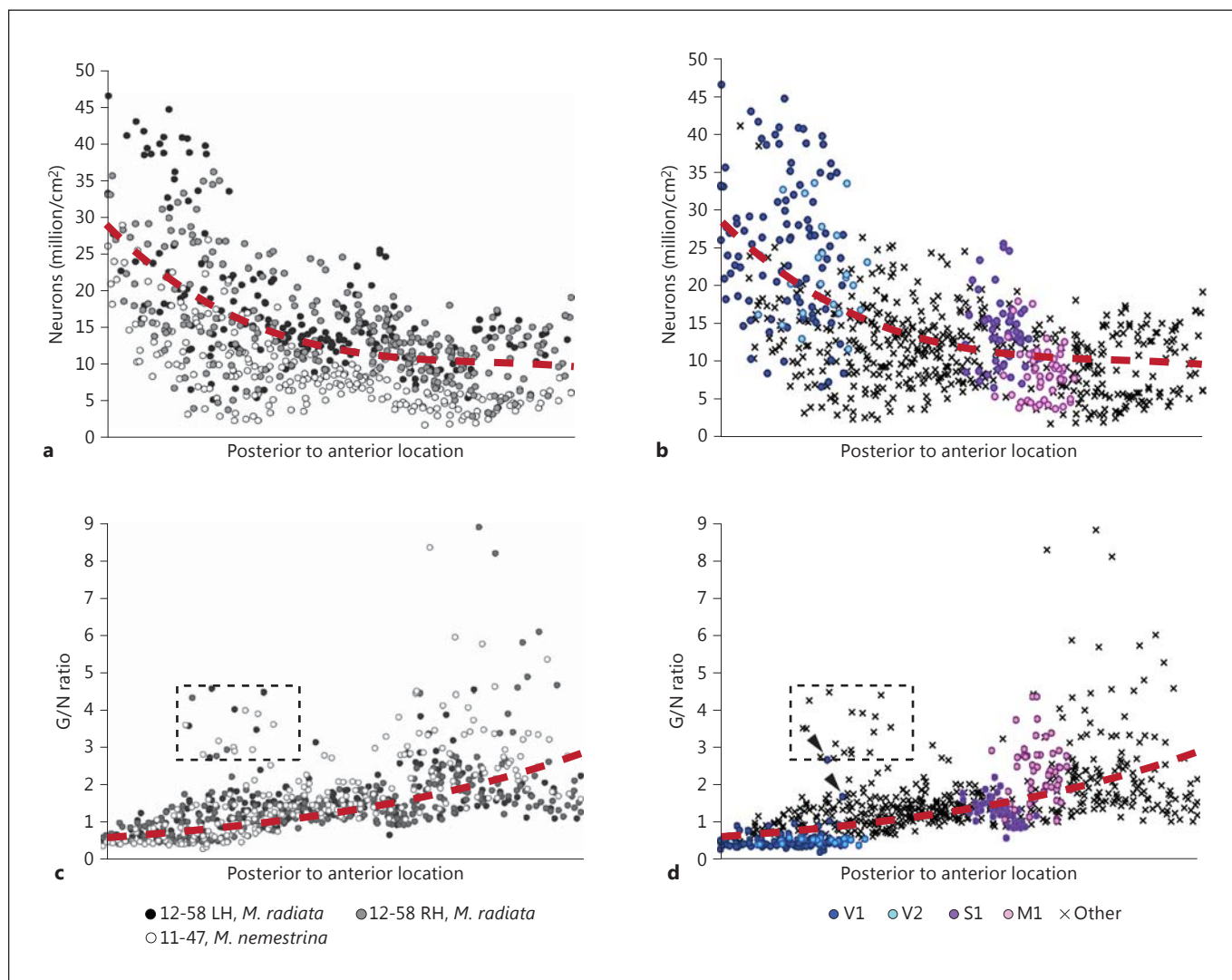
Case 11-47 had a total surface area of 151.8 cm<sup>2</sup> and a total weight of 33.2 g. The hemisphere contained approximately 3 billion cells, of which 1.3 billion were neurons (45%). There was an average of approximately 19.9 million cells/cm<sup>2</sup> and 9.3 million neurons/cm<sup>2</sup> across cortex. V1 was estimated to contain 506 million cells, of which 368 million (73%) were neurons. The cell and neuron densities in V1 were 25.2 million cells/cm<sup>2</sup> and 18.3 million neurons/cm<sup>2</sup>. V2 was estimated to contain 114 million cells, of which 78 million were neurons (68%). V2 contained cell and neuron densities of 24.7 million cells/cm<sup>2</sup> and 16.8 million neurons/cm<sup>2</sup>. We estimated that there were a total of 129 million cells in S1, of which 57 million were neurons (44%). S1 had cell and neuron densities of 19.6 million cells/cm<sup>2</sup> and 8.7 million neurons/cm<sup>2</sup>. M1 was estimated to contain a total 180 million cells, of which 50 million were neurons (28%). The M1 cell and neuron densities were 18.2 million cells/cm<sup>2</sup> and 5.1 million neurons/cm<sup>2</sup>.

Case 10-50 had a total surface area of 152.3 cm<sup>2</sup> and a total weight of 36.4 g. Cell and neuron estimates for this hemisphere are only available for V1, as the rest of the brain was not properly photographed during the dissection. The hemisphere contained 2.9 billion cells, of which 1.4 billion were neurons (48%). There was an average of 19.2 million cells/cm<sup>2</sup> and 9.4 million neurons/cm<sup>2</sup>. V1

was estimated to contain 384 million cells, of which 263 million (68%) were neurons. The cell and neuron densities in V1 were 26.6 million cells/cm<sup>2</sup> and 18.2 million neurons/cm<sup>2</sup>.

#### Neuron Density Varies by Cortical Area

The Kruskal-Wallis test comparing the distributions of both neuron and cell densities in our sample revealed significant differences between different cortical areas (neurons:  $\chi^2 = 121.3$ , d.f. = 3,  $p < 0.0001$ ,  $n = 241$ ; cells:  $\chi^2 = 16.3$ , d.f. = 3,  $p = 0.001$ ,  $n = 241$ ). Most cortical areas were similar in cell density, and the only statistical difference detected was between V1 and M1 ( $Z = 3.94$ ,  $p < 0.001$ ). For neuron density, statistical differences were detected between all cortical areas, with the exception of V1 and V2 ( $Z = 0.35$ ,  $p = 0.728$ ). V1 is the most cell- and neuron-dense area of the cerebral cortex, and the total area ranged between 9.4 and 15% of cortex. Average V1 cell density ranged from 25.2 to 52.6 million cells/cm<sup>2</sup> in comparison to the overall average range of 19.2–37.6 million cells/cm<sup>2</sup>. V1 neuron density ranged from 18.2 to 34.7 million cells/cm<sup>2</sup> in comparison to the overall neuron density range of 9.3–17.7 million cells/cm<sup>2</sup>. V2 neuron densities did not have as many extremely high values as V1, but most values were distributed among V1 values. V2 cell density ranged from 24.7 to 41.1 million cells/cm<sup>2</sup>, and neuron density ranged from 16.8 to 28 million cells/cm<sup>2</sup>. Area S1 contained moderate cell and neuron densi-



**Fig. 4.** **a** Distribution of neuron density across the flattened cortex (cases 12-58 LH, 12-58 RH, and 11-47) in a plot of density versus posterior to anterior location. The red dashed line indicates the curve estimation that most closely correlated with the data ( $y = 2.79 \times 10^7 - 6.25 \times 10^7 x + 7.36 \times 10^7 x^2 - 2.99 \times 10^7 x^3$ ). **b** Distribution of neuron density within each area (V1, V2, S1, and M1) binned across cases in a plot of neuron density versus posterior to anterior location. **c** Distribution of the G/N ratio across each flattened cortex (cases 12-58 LH, 12-58 RH, and 11-47) in a plot of the

G/N ratio versus posterior to anterior location. The red dashed line indicates the curve estimation that most closely correlated with the data ( $y = 0.58^{(1.61x)}$ ). **d** Distribution of the G/N ratio within each cortical area in a plot of G/N ratio versus posterior to anterior location. The arrowheads identify two outlier pieces in the V1 region with higher G/N ratios that may have included some white matter. The dashed boxes in **c** and **d** outline tissue pieces containing a high G/N ratio in posterior cortex, and they correspond to the pieces outlined in figure 2 from piriform cortex.

ties, as S1 is more cell and neuron dense than M1, but less densely packed than V1. S1 cell density ranged from 19.6 to 40.2 million cells/cm<sup>2</sup>, while neuron density ranged from 8.7 to 17.1 million cells/cm<sup>2</sup>. Primary motor cortex is the least neuron-dense area examined, where cell density ranged from 18.2 to 32.9 million cells/cm<sup>2</sup>, and M1 neuron density in macaques ranged from 5.1 to 10.6 mil-

lion cells/cm<sup>2</sup>, which is approximately three times lower than the V1 range.

Neuron density versus the A-P dimension was plotted for each case (fig. 4a), with the exception of case 10-50, whose A-P coordinates were unknown since pieces were not photographed. A similar pattern within the A-P dimension emerges across cases, but there is con-

siderable variation in the range of neuron densities. The neuron distributions follow the same posterior-to-anterior decrease in cortical neuron density that is typical in primates [Collins et al., 2010a, 2016; Cahalane et al., 2012; Young et al., 2013a, b], but average curve fit estimations suggest that the A-P gradient is not simply linear. Moving from posterior to anterior pieces, V1 values are the highest and are mixed with high V2 values, cortical pieces just anterior to these visual areas contain more moderate values, S1 contains pieces with high-to-moderate neuron densities, M1 contains pieces with lower-than-average neuron densities, and the most anterior pieces have moderate values. For neuron densities plotted by A-P location, we found that a cubic model provided the best correlation (adjusted  $R^2 = 0.340$ ) compared to linear (0.279), exponential (0.244), or quadratic models (0.336). However, an exponential model resulted in the lowest root mean square error (RMSE) (0.253), while a linear model resulted in the highest RMSE ( $4.45 \times 10^{13}$ ), indicating the least useful fit for prediction. Figure 4b summarizes the distribution of neurons within each cortical area for all cases combined.

#### *Other Cell/Neuron Ratio Varies by Cortical Area*

To minimize the effects of variability in the raw cell and neuron density measures due to brain sizes in our small sample, we used G/N values to examine differences between cortical areas. The G/N values were binned across cases to examine the ratio in different cortical areas (fig. 3c). Overall, the G/N ratio was consistent within cortical areas across all cases, and there was clear separation between M1, S1, and the two visual areas, V1 and V2, which were very similar. The Kruskal-Wallis test comparing the distributions of the G/N ratio between different cortical areas revealed significant differences ( $\chi^2 = 170$ , d.f. = 3,  $p < 0.001$ ,  $n = 241$ ). However, the pairwise comparison between V1 and V2 was the exception without a statistical difference in the G/N ratio ( $Z = -0.69$ ,  $p = 0.489$ ). The average G/N ratio in V1 ranged from 0.38 to 0.61, while it similarly varied from 0.46 to 0.51 in V2. Two outlier pieces in the V1 region with higher G/N ratios (see arrowheads in fig. 4d) may have included some white matter. The average G/N ratio was doubled from V1 to S1, with average values within cases ranging from 1.19 to 1.39 in S1, and the pairwise comparison supported statistical differences between the distributions ( $Z = -8.31$ ,  $p < 0.001$ ). G/N ratios for V2 versus S1 ( $Z = -5.01$ ,  $p < 0.001$ ) resembled the relationship between V1 and S1. The G/N ratio in M1 was the highest and the range was the most

variable, with the average four times higher than the average V1 ratios ( $Z = -11.75$ ,  $p < 0.001$ ), and values ranging from 1.99 to 2.71. Similarly, the distributions of the G/N ratio in V2 compared to M1 ( $Z = -7.32$ ,  $p < 0.001$ ) resemble the relationships listed for V1, and the pairwise comparison also supported statistical differences between M1 and S1 ( $Z = -2.19$ ,  $p = 0.021$ ). While not included in the statistical analysis, the G/N ratios from the two baboons followed trends by area comparable to the macaques studied. However, the average G/N ratios for the two baboons tended to be slightly lower in each cortical area in comparison to the macaque brains (case 09-27: V1 = 0.37, S1 = 0.86, M1 = 1.90; case 11-31: V1 = 0.40, S1 = 0.69, M1 = 1.53).

The A-P plot of the G/N ratio across each case showed more similar values between cases at each A-P coordinate (fig. 4c) compared to neuron densities across the A-P dimension (fig. 4a). For G/N ratios plotted by A-P location, we found that an exponential model provided the best correlation (adjusted  $R^2 = 0.434$ ) compared to linear (0.276), cubic (0.279), or quadratic models (0.277). The exponential model resulted in the best RMSE (0.224), and the linear model had the largest RMSE (0.744). Figure 4d summarizes the distribution of G/N ratios within each cortical area for all cases combined.

## Discussion

Here we provide estimates of the total number and density of cells and neurons for multiple pieces of tissue across the cortical sheet, and for four specific areas within the macaque cortex. Comparisons are made with previously published data collected from baboons in a comparable way. Our findings are consistent with previous observations that primary visual cortex and early extrastriate visual areas tend to have higher-than-average neuron and cell densities while motor cortex has lower-than-average densities [Collins et al., 2010a, 2011, 2016; Young et al., 2013b]. These results suggest that this distinct architecture exists across most or all anthropoid primates. While the average cell and neuron densities decrease with increasing brain size across cases, the average cellular composition across the brain remains relatively consistent between cases, and the total number of cells and neurons varies little among cases. The calculated G/N ratios provide clear distinctions between cortical areas M1 and S1 versus visual areas V1 and V2 (fig. 3c).

### *The Overall Pattern of Cell and Neuron Densities, and the Sizes of Cortex and Cortical Areas*

In all cases, there was large variation in neuron and cell densities across the many pieces of cortex that were sampled. As the boundaries of the areas were estimates, the densities and especially the values of total size and total number of neurons and cells are not exact and may be underestimates. Likewise, because of small, but variable amounts of missing cortex due to the dissection and flattening process, the values of the total sizes of cortex and total numbers of neurons and cells are likely underestimates. Yet, some of the results were so robust that clear conclusions are possible. First, as previously reported for various anthropoid primates [Collins et al., 2010a; Young et al., 2013b], neuron densities were highest in primary and secondary visual cortexes. Other visual areas near or adjoining V2 also had high neuron densities, indicating that extrastriate visual areas near V1 all have higher than average neuron densities. Primary somatosensory cortex also had higher than average packing of neurons, while primary motor cortex had low values. To some extent, premotor cortex also had low values, as did regions of ventral frontal cortex and the ventral temporal lobe. A region of dorsolateral prefrontal cortex had moderately high neuron densities. As these differences in neuron densities relate to functional roles, it will be important to obtain more precise estimates of neuron numbers relative to areal boundaries and sizes in the future.

The observed variance in neuron densities across our macaque cases may relate in part to differences in brain size (table 2). For example, case 12-58 LH (*M. radiata*), which has the expected lowest cortical weight and cortical surface area (24.8 g, 100.6 cm<sup>2</sup>) [Fooden, 1975, 1981; Schwartz and Kemnitz, 1992], contains the highest cell and neuron densities. Cases 11-47 and 10-50, with 50% greater cortical weights (11-47: 33.2 g; 10-50: 36.4 g) and surface areas (11-47: 151.8 cm<sup>2</sup>; 10-50: 152.3 cm<sup>2</sup>), contain similar total and areal cell and neuron densities to each other (table 2). However, cases 11-47 and 10-50 contain slightly less overall neurons compared to case 12-58. These two factors, the total number of neurons and brain size, may explain the comparatively higher neuron density for case 12-58.

### *Patterns across the A-P Gradient*

All cases examined followed the same pattern of cell and neuron distributions within cortical areas despite our finding that absolute values varied across individuals. Moving from posterior to anterior tissue beginning at V1, the results indicate that (1) V1 has the highest neuron den-

sity; (2) there is a graded decline in neuron density across cortex anterior to V1; (3) somatosensory cortex represents a second peak in neuron density, and (4) just anterior to somatosensory cortex there is another slight decline in neuron density (motor cortex), but then neuron density increases again in the more anterior prefrontal cortical region. The region that represents much of this rise in neuron density is part of prefrontal granular cortex [Preuss and Goldman-Rakic, 1991]. However, the rise in neuron density in this region of prefrontal cortex appears to be more distinct from surrounding areas in the chimpanzee cortex [Collins et al., 2016]. This A-P pattern of neuron density suggests there is a common blueprint of cortical organization between primates, in which there are consistent differences in cell and neuron densities in homologous areas and regions, speculatively due to information-processing demands. An overall A-P gradient of increasing neuron densities has been suggested as a basic pattern derived from a developmental gradient in neuron production [Cahalane et al., 2012], and such a neuron-generating gradient could account for much of the pattern seen in our data. However, the gradual A-P gradient is a result of summing all cortical tissue pieces in the single flattened plane; clear borders emerge between cortical areas when pieces within individual cortical areas are summed. These sharp changes in neuron densities at areal borders, and increased neuron densities in S1 and prefrontal granular cortex, likely depend on factors in addition to the developmental gradients. An overall lateral-to-medial gradient is also a factor to consider, as the developing cortical plate decreases in thickness in the lateral-to-medial direction [O'Leary et al., 2007]. However, it is important to consider the clearly different functional roles of areas with high and low neuron densities, and both the generation and the loss of neurons can influence neuron densities in adults [Rakic et al., 1991; Dehay et al., 2001].

### *Comparisons with Other Species*

The use of the same procedures for flattening cortex and obtaining cell and neuron counts for many small sections of the cortical sheet in several recent studies on primates allows for a number of potentially useful and informative comparisons. First, both average neuron density and the overall proportion of neurons are variable across species. Approximately 47% of all cells were neurons across macaque cortex, which is slightly lower than the overall proportion in baboons (51 and 52%) [Young et al., 2013b] but higher than in chimpanzees (39%) [Collins et al., 2016]. The average cell and neuron densities in macaque cortex (cells: 27.2 million/cm<sup>2</sup>, neurons: 12.8

million/cm<sup>2</sup>) are slightly higher than the average densities reported in baboon (cells: 21.9 million/cm<sup>2</sup>, neurons: 11.3 million/cm<sup>2</sup>) or chimpanzee cortex (cells: 26.2 million/cm<sup>2</sup>, neurons: 10.4 million/cm<sup>2</sup>). However, the larger differences in cell and neuron densities between macaques (cases 12-58, 11-47, and 10-50) suggest that observations on more individuals might reveal more variation.

The slight variability in overall neuron density across species is also apparent in specific cortical areas. In all primates examined, V1 unsurprisingly had the highest neuron densities, and M1 contained the lowest neuron densities [Collins et al., 2010a, 2011; Young et al., 2013b]. The average cell and neuron densities in macaque V1 (cells: 36.1 million/cm<sup>2</sup>, neurons: 24.5 million/cm<sup>2</sup>) are higher than the average densities found in V1 baboon (cells: 27.2 million/cm<sup>2</sup>, neurons: 19.9 million/cm<sup>2</sup>) or chimpanzee cortex (cells: 32 million/cm<sup>2</sup>, neurons: 21 million/cm<sup>2</sup>). Although our estimates of V1 may be an underestimation, the measured surface area of flattened V1 in macaques (9.4–15% of cortex) is consistent with previous studies in baboons (14–15.9%) and chimpanzees (10%).

Furthermore, our finding that the G/N ratios within different cortical areas across all four macaque brains are similar suggests that the variation in packing densities is due to brain size or total number of cells and neurons, and not a counting error. Areas of low neuron density contain larger neurons overall [Friede, 1954; Hawkins and Olzowski, 1957; Tower and Young, 1973] that require more glia per neuron; therefore, areas of low neuron density generally have a higher G/N ratio and the opposite is true for areas of high neuron density. In this respect, the G/N ratio can serve as a useful tool when comparing brains of different sizes (that are closely related) but similar in their overall neuronal composition (e.g., neuron and glial cell size). The average G/N ratio for each macaque brain (case 12-58 LH = 1.40/RH = 1.44, case 11-47 = 1.64, case 10-50 = 1.29) is slightly higher than what has been reported in flattened cortex of baboons (case 09-27 = 1.19, case 11-31 = 1.16) [Young et al., 2013b] and lower than the reported values for one chimpanzee brain (1.86) [Collins et al., 2016]. The average G/N ratios in macaque V1 (range = 0.38–0.61) is between the average G/N ratios described in V1 of baboons (0.37, 0.40) and a chimpanzee (0.58). The G/N ratios in macaque areas S1 (range = 1.19–1.39) and M1 (range = 1.99–2.71) are slightly higher than those reported in two baboons (S1 = 0.86–0.69; M1 = 1.90–1.53) and more similar to what has been described in chimpanzee neocortex (S1 = 1.25, M1 = 2.96). However, it is unclear if the G/N ratios here reveal any true

differences between these primate species. S1 and M1 are more challenging than V1 to identify without electrophysiological data, and it is likely for each species that the M1 region includes some cortex from area 3a and the S1 region includes cortex from areas 3a, 1, and 2. There are differences in brain size between macaques, baboons, and chimpanzees, but assuming there are no significant changes in average neuron size [Harrison et al., 2002], we would not expect the G/N ratio to vary much.

### *Nonuniformity across the Cortical Sheet*

The results of a highly influential study three decades ago led to the conclusion that neuron densities did not vary much across cortical areas and species, with the exception that neuron density doubled in V1 of macaque monkeys [Rockel et al., 1980; Carlo and Stevens, 2013]. Such findings may have been the result of undersampling [Cahalane et al., 2012; Herculano-Houzel et al., 2013]. Our estimates of neuron numbers using flow and isotropic fractionation from the current study and previous research in prosimians, other Old World monkeys, and hominins [Collins et al., 2010a, 2011, 2016; Young et al., 2013b] have provided evidence for fluctuations in neuron density in areas anterior to V1. The current methodology has proven to produce results that are not significantly different from those of other methodologies used to count cell and neuron numbers [Bahney and von Bartheld, 2014; Miller et al., 2014], and these ongoing studies provide many more estimates of neuron densities from across the entire cortical sheet. It has been previously reported in other primate species using this methodology that M1 contains lower neuron densities [Young et al., 2013a], and studies on rodents and cats have also reported that M1 contains lower neuron densities than other areas of cortex [Beaulieu and Colonnier, 1989; Allman, 1994; Skoglund et al., 1996; Young et al., 2013a]. In the present study, we found that the number of neurons beneath 1 mm<sup>2</sup> of cortical surface is variable between cortical areas across cases. As first reported three decades ago, primary visual areas have the highest densities of neurons, but secondary visual areas, somatosensory areas, and some of prefrontal cortex also contain higher-than-average neuron densities, while motor areas contain lower-than-average neuron densities. Similarly, the G/N ratio range is significantly different between some cortical areas, suggesting that the number as well as the size of neuronal and glial cells, and therefore density, differs depending on the functional requirements of that particular cortical area.

### Functional Implications

Neuron density and average neuron size are inversely related [Herculano-Houzel, 2014; Herculano-Houzel et al., 2014], as larger neurons take up more space and are surrounded by more glial cells which do not vary much in size. Thus, as neuron densities decrease, glial cell densities increase. This inverse relationship is reflected in the G/N ratio that is high in motor cortex and low in V1. These differences likely reflect the different functional roles of neurons in M1, V1, and other cortical areas. Smaller neurons, such as granule cells in layer 4 of V1, are thought to preserve information provided by only a few driving inputs and serve to accurately relay this information from the lateral geniculate nucleus to other cortical areas [Elston et al., 1996; Kaas, 2000; Srinivasan et al., 2015]. Larger neurons, such as Betz cells in layer 5 of motor cortex, require more glial support, have larger dendritic arbors [Elston, 2002], and integrate information from many sources. The G/N ratio increases in areas where there are fewer, but larger neurons, and this is seen in areas such as M1, which in macaques contains on aver-

age a G/N ratio that is four times higher compared to V1. Other cortical areas vary in neuron densities, and thereby average neuron size, and this variation likely reflects their differing functional roles.

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The authors have no conflict of interest to disclose.

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